

RIYADH

HeatReady

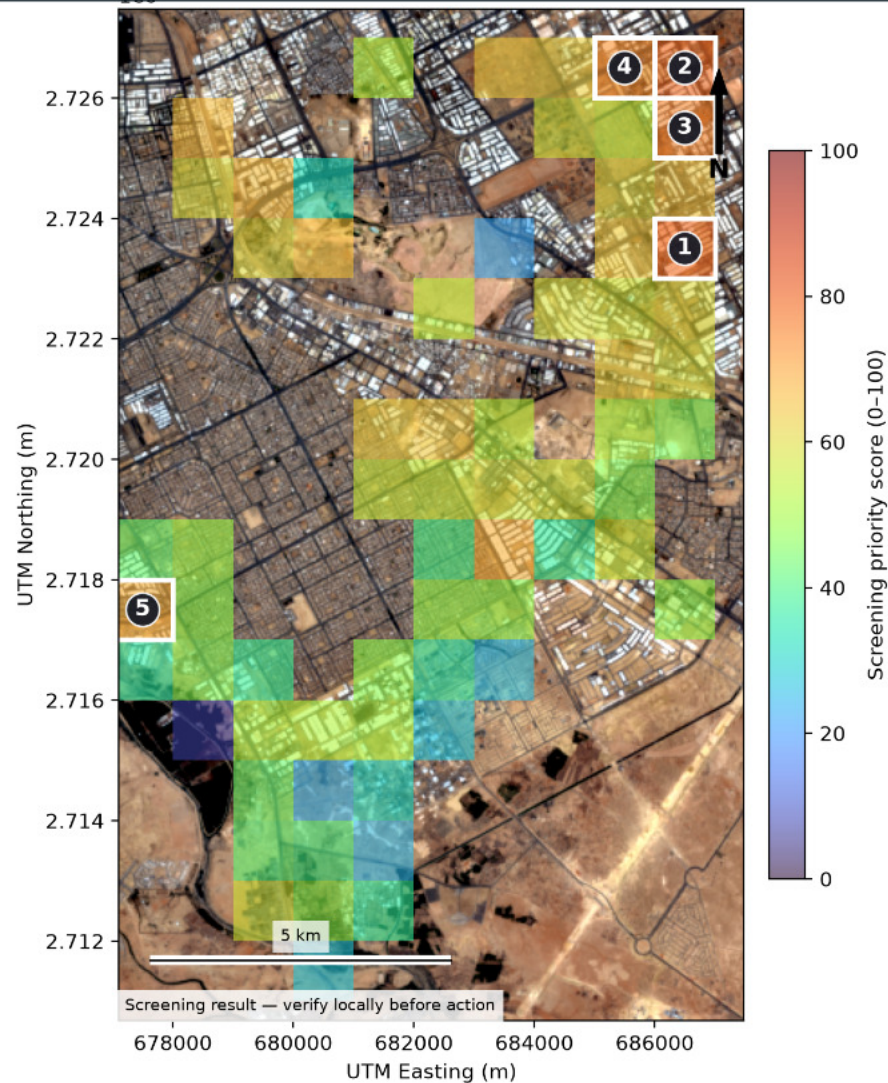
A growth-to-heat priority map for
southeastern Riyadh

One clear decision

**Which 1 km zones should a municipal
planning and heat-resilience team
inspect first?**

Milkaholics • Urban Expansion, Land Use Change & Heat Risk • United
Arab Emirates

SATELLITE-DERIVED PRIORITY MAP



Growth and heat are visible - but the decision is fragmented

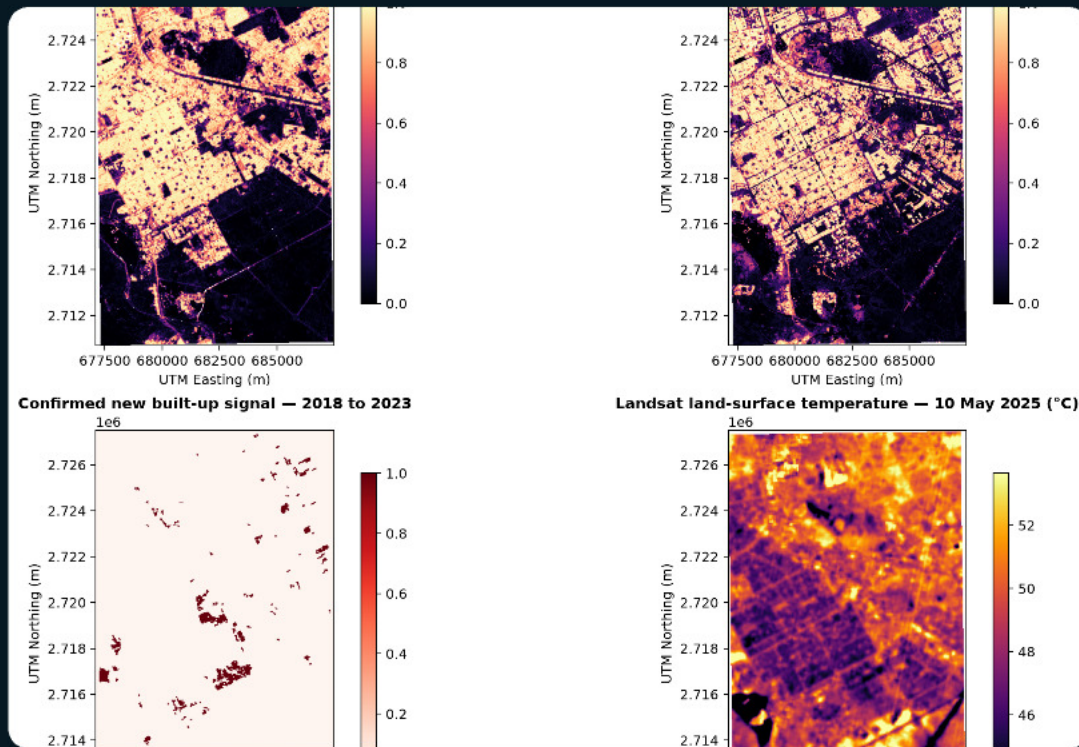
168.30

km² study
area

Planners may need to compare
separate growth, heat, population,
and vegetation layers before
choosing where to inspect.

Affected group

Residents and workers in
fast-changing, hot urban-fringe zones.



One user. One decision. One testable PoC.

USER

Municipal planning + heat-resilience team

Reviews zones, local plans,
infrastructure, and field
conditions.



DECISION

Where should we inspect first?

Output: ranked 1 km cells with
the evidence behind each
score.



ACTION

Local review before intervention

Shade, trees, cool roofs, or
pavements are options to assess
- not automatic recommendations.

HONEST GAP

No municipal user interview has happened yet. The current workflow and value are hypotheses to test during incubation.

Every source has one job - and the result depends on EO

Sentinel-2

Optical reflectance, quality mask, vegetation + built features

Sentinel-1

Radar structure to reduce bright-desert confusion

Annual LULC

Six mapped years propose 2018-2023 growth

Landsat 8

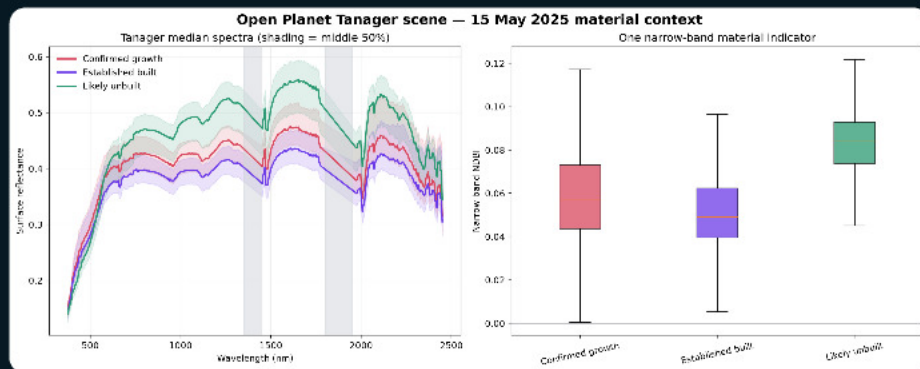
2025 land-surface temperature

WorldPop

Modelled 2025 population exposure

Planet Tanager

426-band 2025 spectra: material context + failure-mode test



Hyperspectral role

Tanager improves the analysis by testing a known desert failure mode. Likely-unbuilt surfaces had higher median narrow-band NDBI than established-built surfaces.

A conservative chain from open scenes to inspection priority

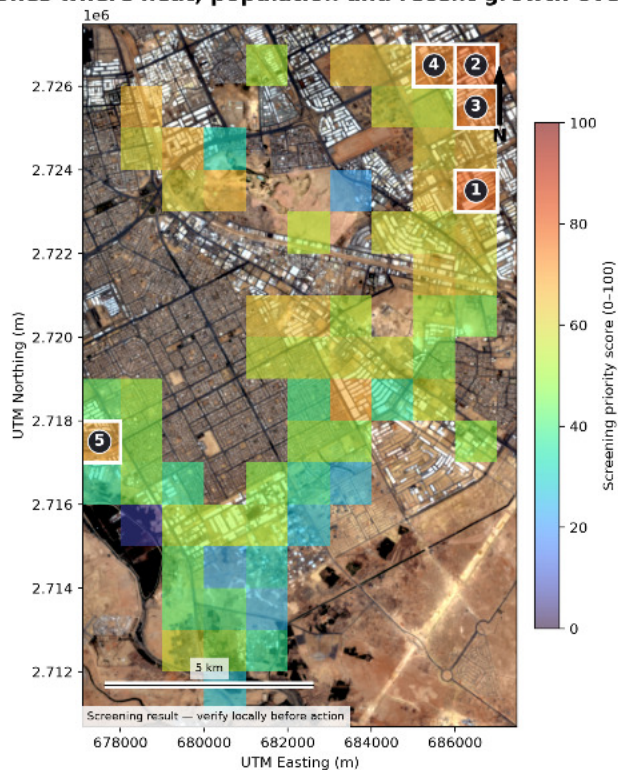


Validation design

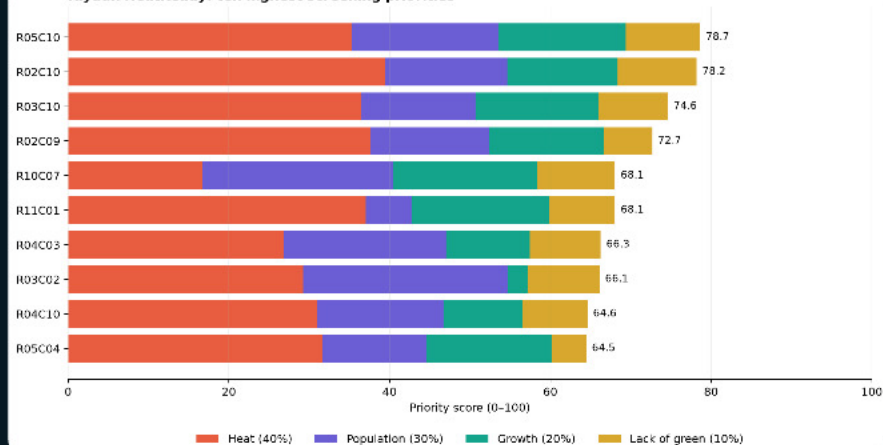
The east side is held out for testing against WorldCover. The final growth layer must pass two independent tests: annual change + optical-radar confirmation.

Six data sources become one ranked inspection list

Riyadh HeatReady decision map
Top zones where heat, population and recent growth overlap



Riyadh HeatReady: ten highest screening priorities



3.74 km²

confirmed growth
signal

14 / 67

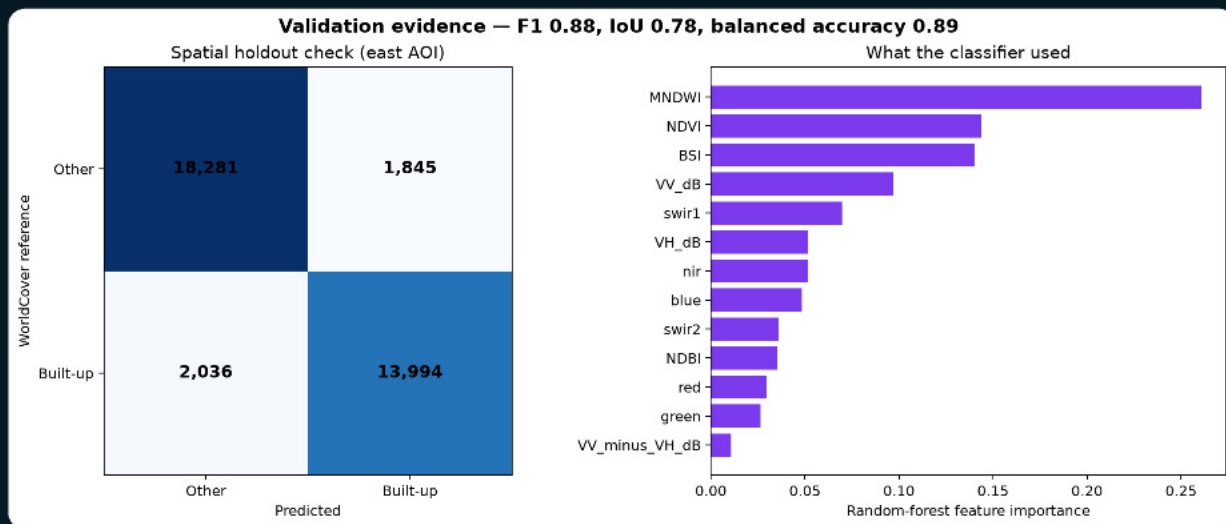
High / eligible cells

78.7

top score: R05C10

Screening result - verify locally before action

Strong satellite agreement - with clear boundaries on the claim

**0.88**

F1 agreement

0.78

intersection over union

Four limits to remember

- 1 WorldCover agreement is not field accuracy.
- 2 Surface temperature is not air temperature.
- 3 WorldPop is modelled, not a census.
- 4 Bright desert can still confuse built surfaces.

The 2022 mapped-area dip is useful evidence of uncertainty - not something to hide.

A repeatable screen can focus scarce local inspection time

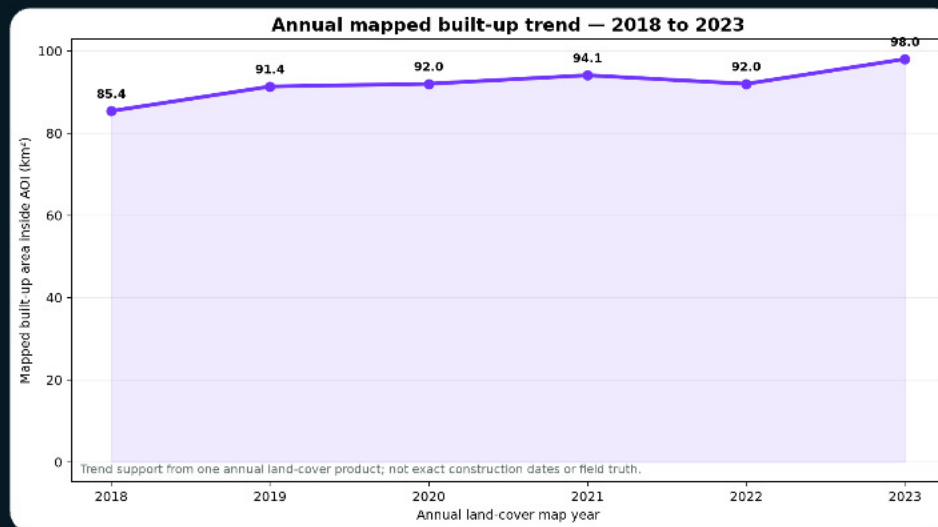
PILOT HYPOTHESIS

Inspect the top five zones

A planning team compares the ranking with local plans and field conditions, then records useful hits, false alarms, and time saved.

Deliverable

**Refreshable dashboard + CSV
+ GeoJSON + GeoTIFF**



85.39

km² mapped in 2018

98.03

km² mapped in 2023

1 area

tested; not all Riyadh

Turn a credible screen into a user-validated municipal tool

- 01 Interview** 3+ planners, GIS staff, or heat stakeholders
- 02 Agree** User decision, weights, thresholds, and success measure
- 03 Validate** Field-check top zones and label false alarms
- 04 Enrich** City plans, worker exposure, and higher-resolution data
- 05 Repeat** Run another Riyadh area and test update cost

Goal: prove usefulness with real users - then scale the data and product.

Data traceability and licence record

CHANGE AND BUILT SIGNAL

Sentinel-2 L2A

Copernicus open
access

11 May 2018, 10 May 2021, 9 May 2025

Sentinel-1 RTC

CC BY 4.0

10 May 2018, 6 May 2021, 9 May 2025

Annual LULC

CC BY 4.0

Impact Observatory / Esri, 2018 to 2023

Built reference

CC BY 4.0

ESA WorldCover 2021, class 50

HEAT, EXPOSURE AND SPECTRA

Landsat 8 C2 L2

LC08_L2SP_166043_20250510_02_T1

10 May 2025, US public domain

WorldPop Global2

sau_pop_2025_CN_1km_R2025A_UA_v1.tif

2025 alpha release, CC BY 4.0

Planet Tanager

20250515_080954_16_4001

15 May 2025, CC BY 4.0

Full item IDs, providers, processing roles and licence links
are recorded in `config.json` and
`docs/DATA_AND_LICENSES.md`.

Validation and conservative change logic

SPATIAL MODEL CHECK

WEST

Train

MIDDLE

Tune
threshold

EAST

Test once

Gaps separate the three areas to reduce spatial leakage

0.88

F1 agreement

0.78

intersection over union

0.89

balanced accuracy

36,156 east-side test pixels

These metrics show agreement with WorldCover. Field accuracy still requires local reference data.

GROWTH CONFIRMATION

13.82 km²

annual map candidate

2018 non-built to 2023 built



Three confirmation rules

2025 optical-radar model says built
Probability gain is at least 0.10
Connected group has at least six pixels



3.74 km² retained as confirmed growth signal

Judge questions and evidence-based answers

Why these weights?

They are transparent PoC assumptions. A municipal pilot should review heat, population, growth and vegetation weights.

What does 0.88 mean?

F1 agreement with WorldCover on an east-side holdout. Independent field accuracy remains unmeasured.

Why use Tanager?

Its narrow bands test a desert failure mode. Likely-unbuilt land had higher median NDBI than established built land.

Can judges rerun it?

The committed notebook rebuilds the 67-cell ranking in under one minute. The full pipeline records every scene and parameter.

First real pilot: inspect the top five zones with one planning team and record useful hits, false alarms and workflow fit.